

Seed Annotation Guidelines

1. Purpose and Scope

This document defines annotation rules for seed images collected by our collaborator from the University of Gothenburg. These guidelines are developed specifically for this thesis project and the provided dataset, and are not intended to generalize to other seed datasets or species.

Annotations are created using CVAT (Computer Vision Annotation Tool) on a self-hosted server to be able to collaborate together.

Inter-annotator agreement is assessed using Cohen's kappa on selected subsets.

2. Species Priority Order

- Annotate in the following order in DSR cycle 1
 1. VAU
 2. PHYCA
 3. PEH
 4. CAT

3. Annotation Classes

Use exactly two bounding box classes. Do not create additional classes.

Class	CVAT Tag	Definition
Seed	seed	Seed with characteristic color and shape for the species. Including both mature and immature ones.
Unsure	unsure	Cannot determine seed/debris. A domain expert will review and resolve all unsure labels before export.

Do not annotate objects that are clearly debris. If it is unclear whether the object is a seed or debris, annotate it as "unsure" for later review.

4. Species-Specific Rules

- **PHYCA**
 - Mature seeds have visible ridges, use ridges to distinguish from debris when unsure.
- **VAU**
 - Zoomed-in images are preferred for annotation.

- Avoid annotating images that are too zoomed in, excessive overlap makes individual seeds hard to distinguish.
- **PEH**
 - In very zoomed-out images: trash vs seed can be differentiated by shape.
 - All PEH seeds are a dark blackish color
 - Use the more zoomed-in PEH images when possible, as seeds vs trash are easier to distinguish.
- **CAT**
 - Follow general rules. No additional species-specific rules established yet.

5. Bounding Box Rules

- Draw rectangle bounding boxes only (they can be rotated). Do not use polygons or keypoints.
- The box should be tight around the visible seed — do not include large amounts of background.
- If a seed is partially covered but clearly identifiable, annotate it using the visible portion.
- If two seeds overlap so much that they cannot be separated, annotate each individually with overlapping boxes.
- Do not annotate the ruler/scale bar at the bottom of the image.
- Do not annotate handwritten labels, a separate OCR step will handle these.

6. Annotation Workflow

- Each seed species has a project (e.g “PHYCA-seeds”), all uploads are tied to a task (e.g. “PHYCA-20260410”).
- Each task has one job per image so that several collaborators can work within the same task (but only one per job).
- Assign the job to you.
- Annotate all visible and identifiable seeds using the two classes above in each image.
- If unsure about a seed, label it unsure, do not guess.
- Submit the annotation when finished with an image.
- Domain expert reviews all unsure labels and a random sample of seed labels for quality control.
- Inter-annotator agreement (Cohen's kappa) is computed on a subset of each species.
- Before exporting, ensure all unsure labels have been resolved or removed. No unsure labels are allowed in the final dataset.
- Export format: Ultralytics YOLO Oriented Bounding Boxes 1.0

7. Image Quality Rules

Prefer zoomed-in images over zoomed-out images when both are available for the same sample.

Skip images if:

- The image is too blurry to distinguish individual seeds
- Excessive overlap makes it impossible to separate seeds or annotate
- Seeds cannot be reliably distinguished from debris

For questions about specific cases, contact our domain expert or post in the group channel for discussion.